

Subsystem	Component Name	Technical Specifications	Selection Rationale & Strategic Value
Main Controller	Teensy 4.0	600 MHz ARM Cortex-M7 processor	Delivers over 30x the processing speed of standard Arduino boards, allowing microsecond sensor filtering and rapid execution of search-and-attack logic.
Drive Motors	4x 32GP Planetary DC Gear Motors	12V DC, 160 RPM, 32mm outer diameter, planetary gearbox	Planetary gearboxes distribute load pressure across three internal gears to survive heavy kinetic impacts; 160 RPM balances maximum torque to push 2.5 kg opponents with line-crossing speeds (<1.5 seconds).
Motor Drivers	BTS7960 H-Bridge Drivers	43A continuous high-current capacity, large passive aluminum heatsink	Prevents driver burnout under sustained motor stall currents during head-on pushing matches where standard L298N drivers fail.
Wheel Couplers	6mm Brass Hex Couplings	Heavy-duty brass construction with set screws	Locks 70mm wheels tightly onto the 6mm motor D-shafts, eliminating rotational slippage during maximum torque exertion.
Vision & Opponent Detection	TOF400C / VL53L1X Time-of-Flight Sensors	Laser-based photon flight-time measurement, up to 4-meter range	Calculates distance via light return time rather than reflection intensity, ensuring reliable detection of dark or matte-black opponent chassis regardless of color.

Line & Ring Edge Detection	TCRT5000 / QRE1113 Analog Line Trackers	Micro ground IR reflectance sensors, bottom-mounted beneath blade and chassis	Continuously monitors the floor surface to detect the white outer ring boundary, triggering immediate motor braking, reversing, and turn-away routines.
Power Supply	3-Cell (3S) LiPo Battery	11.1V nominal (12.6V max), 1800–2200 mAh capacity, 35C–50C discharge rating	Complies strictly with competition maximum 12V power rules; high C-rating provides immediate high-current bursts to all 4 motors without dropping logic rail voltage.
Voltage Regulator	LM2596 DC-DC Buck Converter	Step-down switching regulator (5V / 3.3V outputs)	Steps down 12V battery power to supply clean 5V and 3.3V logic levels to the Teensy and laser sensors, protecting sensitive chips from motor electrical noise.
Traction & Drive Wheels	Compressed Foam Tires	70mm outer diameter, 30–40mm tread width, hollow rims	High-density foam tread maximizes friction on the ring surface while absorbing violent impacts to prevent bending the 6mm motor shafts.

Team Structure and Categories

- Teams consist of two students and one coach. The coach provides organizational guidance, but students must independently build and program the robot.
- The competition is divided into Junior (ages 6–12), Senior (ages 13–19), and University (ages 18–22) divisions.

Robot Physical Constraints

- Senior category robots must fit strictly within a 20x20 cm square footprint. Height is unlimited.
- The maximum allowable mass for the robot is 2.5 kg.

Electrical and Motor Regulations

- Power must come from a single battery providing a maximum of 12V. Boost converters (step-up voltage regulators) are strictly forbidden.
- Actuators are restricted to standard brushed DC motors with a maximum speed rating of 500. Brushless motors (BLDC), linear actuators, and electromagnets are banned.
- Senior teams are permitted to use custom microcontrollers (such as Arduino), 3D-printed parts, and digital fabrication. Pre-assembled commercial sumo kits are strictly prohibited.

Prohibited Features

- Robots cannot use suction devices, magnets, adhesives, sticky tires, flipping mechanisms, offensive weapons, sharp edges, or white sheets designed to disrupt opponent optical sensors.

Match Format and Arena

- Matches are held on a circular ring with a total diameter of 120 cm, featuring a 115 cm black inner playing area surrounded by a 5 cm white border line.
- Matches follow a best-of-three rounds format with a maximum total match time of 3 minutes.
- Robots must feature a mandatory 5-second delay after the start button is pressed before any motors can engage.
- The primary objective is to push the opponent completely out of the ring using kinetic force and autonomous positioning.